

F1

$$m = 0.15 \text{ kg}$$

$$k = 1900 \frac{N}{m}$$

$$x_o = 0.045 \text{ m}$$

$$V_{piso} = ?$$

$$E_o = E_f$$

$$\frac{1}{2} * k * x_o^2 = \frac{1}{2} * m * V_A^2$$

$$V_A = \sqrt{\frac{k * x_o^2}{m}} = \sqrt{\frac{1900 * 0.045^2}{0.15}}$$

$$V_A = 5.06 \text{ m/s}$$

$$y_f = y_o + V_{oy} * t - \frac{1}{2} * g * t^2$$

$$0 = 1.2 - \frac{1}{2} * (9.8) * t^2$$

$$t = \sqrt{\frac{1.2 * 2}{9.8}}$$

$$t = 0.49 \text{ seg}$$

$$V_{fy} = V_{oy} - g * t$$

$$V_{fy} = -9.8 * 0.49$$

$$V_{fy} = -4.80 \text{ m/s}$$

$$V_f = \sqrt{V_{fx}^2 + V_{fy}^2}$$

$$V_f = \sqrt{5.06^2 + 4.80^2}$$

$$V_f = 6.97 \text{ m/s}$$

F2

$$\begin{aligned}E_{co}+E_{po}+E_{ro}+W_{otras}&=E_{cf}+E_{pf}+E_{rf} \\(1/2)(2000)(4)^2-(17000)(2)&=(2000)(10)(-2)+(1/2)k(-2)^2 \\16000-34000&=-40000+2k \\2k&=22000 \\k&=11000 \text{ N/m}\end{aligned}$$

F3

$$mg=qE$$

$$E = \frac{mg}{q} = \frac{10 * 4 \times 10^{-13}}{2.4 \times 10^{-18}} = 1.66 \times 10^6$$

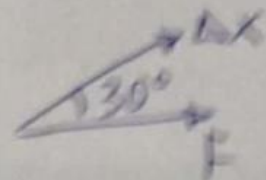
Para el sistema de placas paralelas

$$V=Ed=(1.66 \times 10^6)(0.020) = 33kV$$

F4

$$W_{\text{TOTAL}} = W_F + W_{F_g}$$

$$W_F = F \Delta x \cos 30$$



$$W_F = 40\sqrt{3} (10) \frac{\sqrt{3}}{2}$$

$$W_F = 600 \text{ J}$$

$$W_{F_g} = F_g \Delta x \cos 120$$

$$= 10(10)(10)(-0.5)$$

$$W_{F_g} = -500 \text{ J}$$

$$\therefore \boxed{W_T = 100 \text{ J}}$$

F5

②

posibilidad
bloque baja

$$F_x = F \cos \theta$$
$$F_x = 200 \cos 37^\circ$$
$$F_x = 200 \left(\frac{4}{5} \right) = 160 \text{ N}$$
$$F_y = F \sin \theta = 120 \text{ N}$$
$$mg_x = mg \sin \theta = 300 \text{ N}$$
$$mg_y = mg \cos \theta = 400 \text{ N}$$
$$N = F_y + mg_y \quad N = 520 \text{ (N)}$$
$$F_x - F_f - mg_x = ma$$

~~mgx~~

$$mg_x - F_x - F_f = ma$$
$$300 - 160 - 0.2(520) = 50 a$$
$$\boxed{\frac{18}{25} = a}$$